

Worksheet – 3

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Aim/Overview of the practical: Create a table named **STUDENT** with the following columns: S_id (Primary Key), S_name, Course, Marks, City, and then demonstrate the use of **DDL, DML, DQL, DCL, and TCL** commands.

ApparatusUsed:

- MySQL(Workbench)

Procedure/Algorithm/Code:

1. DDL(Data Definition Language)

The IF statement in Microsoft Excel is a logical function that checks whether a condition is TRUE or FALSE. It returns one value if the condition is true and another value if the condition is false.

SQL Query:

Create EMPLOYEE table

```
CREATE TABLE STUDENT (  
    S_id INT PRIMARY KEY,  
    S_name VARCHAR(50),  
    Course VARCHAR(50),  
    Marks INT,  
    City VARCHAR(50)  
);
```

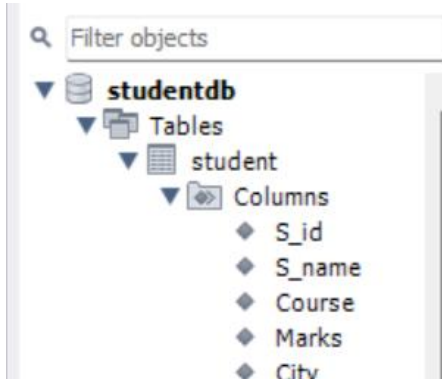
ALTER — Add a new column

```
ALTER TABLE STUDENT  
ADD Email VARCHAR(100);
```

Remove the column

```
ALTER TABLE STUDENT  
DROP COLUMN Email;
```

OUTPUT:



2. DML(Data Manipulation Language)

A Nested IF statement is an IF function placed inside another IF function. It is used to test multiple conditions and return different results based on the first condition that evaluates to TRUE.

SQL Query:

INSERT:

INSERT INTO EMPLOYEE

INSERT INTO STUDENT

VALUES

(101, 'Rahul', 'MCA', 85, 'Chandigarh'),

(102, 'Priya', 'MBA', 78, 'Delhi'),

(103, 'Aman', 'MCA', 92, 'Mohali'),

(104, 'Neha', 'BCA', 88, 'Chandigarh'),

(105, 'Rohit', 'BBA', 75, 'Delhi'),

(106, 'Anjali', 'MCA', 95, 'Mohali');

UPDATE:

UPDATE STUDENT

SET Marks = 90

WHERE S_id = 102;

DELETE:

DELETE FROM STUDENT

WHERE S_id = 105;

3. DQL(Data Query Language)

The AND function checks multiple conditions and returns TRUE only if all the conditions are TRUE. If any one condition is FALSE, it returns FALSE.

SQL Query:

SELECT S_name, Course, Marks

FROM STUDENT

WHERE Marks > 85;

OUTPUT:

Result Grid Filter Rows: Ed					
	S_id	S_name	Course	Marks	City
▶	101	Rahul	MCA	85	Chandigarh
	102	Priya	MBA	78	Delhi
	103	Aman	MCA	92	Mohali
	104	Neha	BCA	88	Chandigarh
	105	Rohit	BBA	75	Delhi
STUDENT 1 x					

4. DCL(Data Control Language)

The OR function checks multiple conditions and returns TRUE if at least one of the conditions is TRUE. It returns FALSE only when all the conditions are FALSE.

SQL Query:

GRANT:

```
GRANT SELECT, INSERT
ON STUDENT
TO user1;
```

REVOKE:

```
REVOKE INSERT
ON STUDENT
FROM user1;
```

Result Grid Filter Rows: Export:	
Grants for user1@localhost	
▶	GRANT USAGE ON *.* TO 'user1'@'localhost'
	GRANT SELECT ON 'studentdb'. 'student' TO ...

5. TCL(Transaction Control Language)

The NOT function reverses the result of a logical condition. It returns TRUE if the condition is FALSE, and FALSE if the condition is TRUE.

SQL Query:

```
UPDATE and COMMIT: UPDATE
STUDENT
SET Marks = 88
WHERE S_id = 104;

COMMIT;COMMIT;
```

ROLLBACK:

```
UPDATE STUDENT  
SET Marks = 50  
WHERE S_id = 103;
```

ROLLBACK;

SAVEPOINT:

```
SAVEPOINT sp1;
```

FINAL TABLE:

Result Grid




Filter Rows:

Edit:





	S_id	S_name	Course	Marks	City
▶	101	Rahul	MCA	85	Chandigarh
	102	Priya	MBA	90	Delhi
	103	Aman	MCA	50	Mohali
	104	Neha	BCA	88	Chandigarh
	106	Anjali	MCA	95	Mohali
✱	NULL	NULL	NULL	NULL	NULL



Learning outcomes:

- Understood the purpose and use of DDL, DML, DQL, DCL, and TCL commands in SQL.
- Created and managed an EMPLOYEE table using SQL database commands.
- Applied INSERT, UPDATE, DELETE, and SELECT commands to manage and retrieve employee data.
- Developed an understanding of database access control using GRANT and REVOKE commands.
- Learned to manage database transactions using COMMIT, ROLLBACK, and SAVEPOINT commands.
- Improved practical skills in creating, manipulating, querying, and controlling data in a relational database.